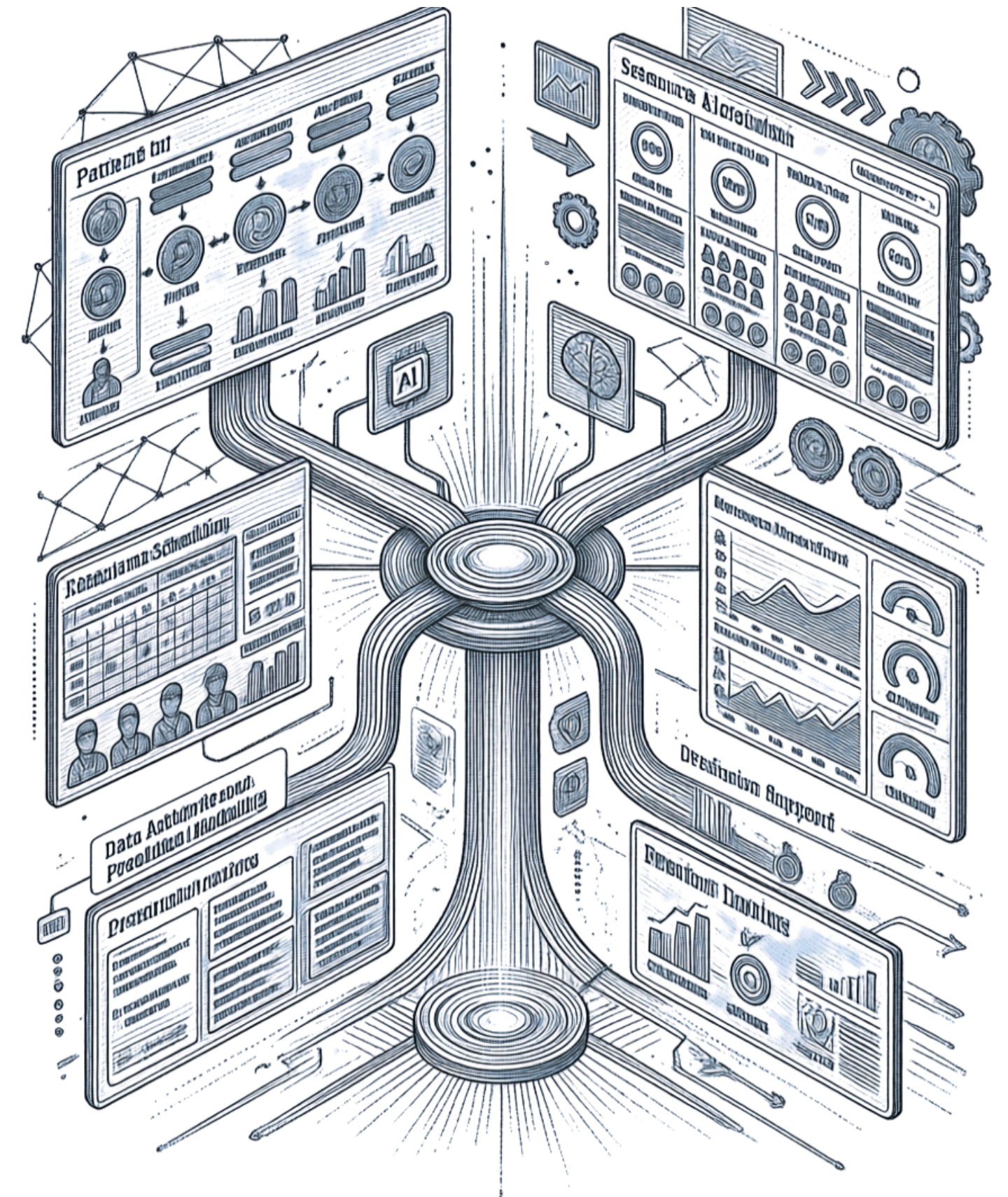


DEVELOPMENT OF A Healthcare Operations Intelligence Dashboard with Decision Analytics

Project Focus

Healthcare data management • Operational monitoring • Interactive analytics
Decision support • Forecasting • Alerts

GROUP 01 | Mentor: Ms. Mousami Shrivastava



THE HEALTHCARE DATA CHALLENGE

Why hospital data needs to become actionable intelligence

PROBLEM STATEMENT

- Hospital information is spread across different systems and files.
- Staff do not have one clear view of waiting time, bed occupancy, emergencies and resources.
- Manual analysis takes time and can delay important operational decisions.
- Operational bottlenecks may be noticed only after they affect service quality.

OUR SOLUTION

- Bring hospital data together in one Healthcare Operations Intelligence Dashboard.
- Clean and organize the data for reliable KPI and trend analysis.
- Monitor patient flow, capacity, emergencies and resource performance in one place.
- Use decision analytics, alerts and forecasting-ready insights to support faster action.

Fragmented data → Centralized dashboard → Clear insights → Proactive decisions

DASHBOARD CAPABILITIES & USER VALUE

Turning raw hospital data into an understandable operational picture

KPI Monitoring

Appointments, emergency cases, patient operations, laboratory activity and operational KPIs.

Interactive Analysis

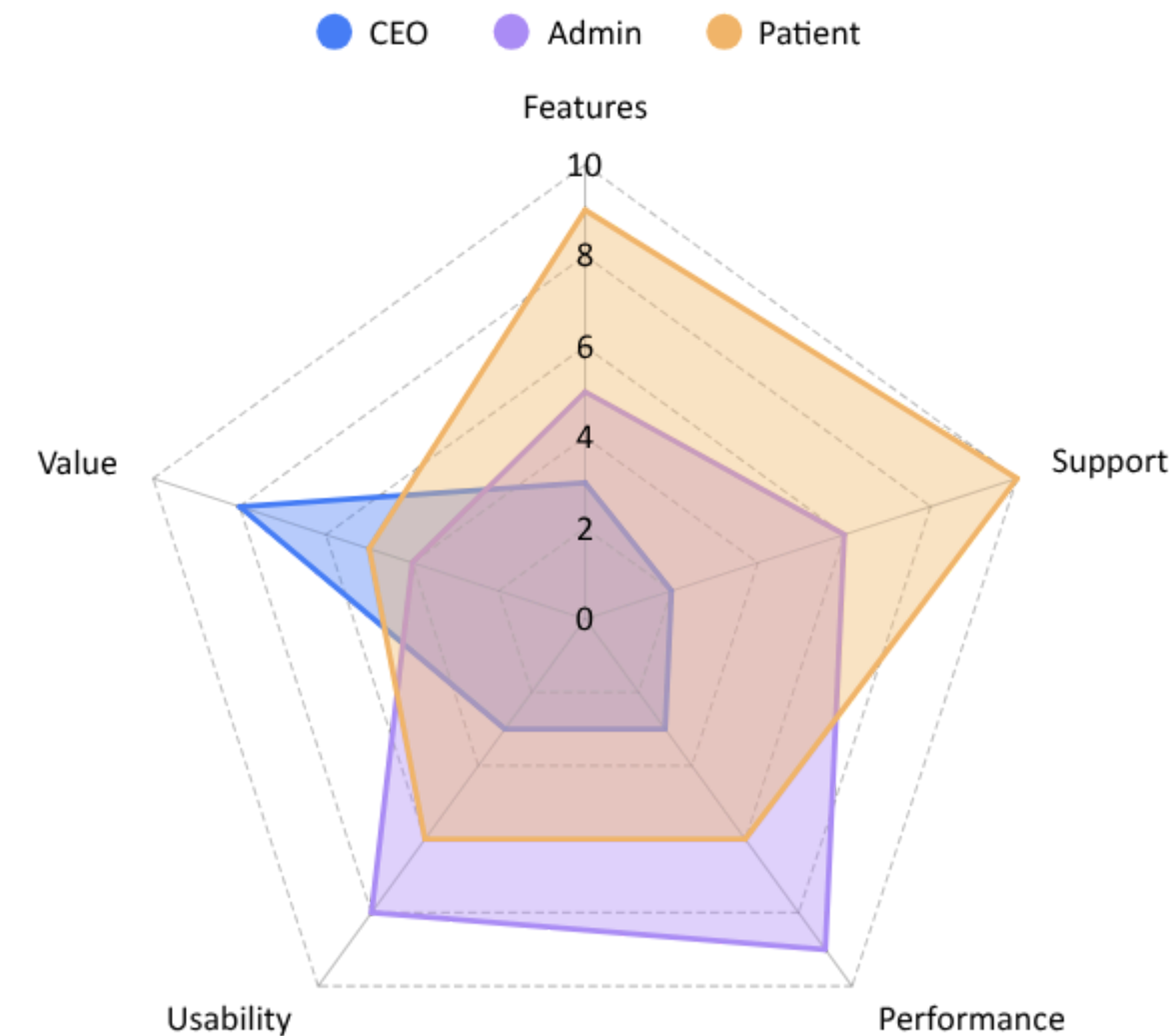
Filters, date ranges and department/category views for drill-down analysis.

Data Visualization

Bar, line and pie charts plus trends and correlations for rapid interpretation.

Decision Support

Critical-condition detection, resource-planning signals and forecasting-ready insights.



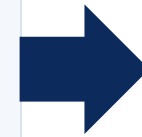
CEO / EXECUTIVE Strategic KPIs & trends | ADMIN Detailed operations & resources | PUBLIC Selected general information

SYSTEM ARCHITECTURE & TECHNOLOGY STACK

Development flow: data foundation → analytics engine → intelligence delivery

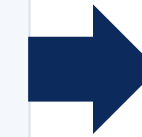
1 DATA FOUNDATION

Excel / Hospital datasets
OpenPyXL + Pandas
Clean • transform • validate



2 ANALYTICS ENGINE

Python + Pandas
KPI calculation • trends • correlations
Statsmodels forecasting



3 INTELLIGENCE DELIVERY

Streamlit dashboard
Decision analytics • alerts • forecasting
Pytest validation

COLLECT → CLEAN → TRANSFORM → ANALYZE → PREDICT → ALERT → DECIDE

Raw hospital data → reliable data layer → operational intelligence → decision support

DECISION ANALYTICS: FROM SIGNALS TO ACTION

Moving beyond “what is happening?” toward “what requires attention?”

High waiting time

Review patient flow; increase counters/staff when needed.

High bed occupancy

Plan additional capacity and accelerate discharge.

Emergency overload

Deploy additional triage/support resources.

Low ambulance response

Review routing and fleet allocation.

High OT utilization

Optimize scheduling and capacity.

KPI / Operational Signal

Compare with threshold

Condition requires attention?

Generate alert / insight

Management reviews

Take appropriate action

FUTURE SCOPE & SCALE-UP ROADMAP

From an analytics dashboard to a connected healthcare intelligence layer

01

Real-Time Data Integration

Live hospital databases & APIs

02

AI-Based Prediction

Patient, emergency, bed, staff & pharmacy demand

03

Automated Notifications

Email, SMS & mobile threshold alerts

04

Smart Resource Allocation

Beds, staff, ambulances & operating theatres

05

Mobile & Secure Access

Role-based mobile access & stronger controls

06

Hospital System Integration

Connect with existing hospital management systems

Expected Impact • Faster visibility • Earlier intervention • Better resource utilization • Proactive planning • Scalable operations

CONCLUSION & EXPECTED OUTCOMES

The project's north star: convert hospital data into decision-ready operational intelligence

✓ Centralized view of hospital operations

✓ Critical KPI and trend monitoring

✓ Earlier identification of bottlenecks

✓ Interactive, role-based decision support

✓ Resource and capacity planning

✓ Alerts and forecasting for proactive action

✓ Shift from reactive reporting to data-driven management

HOSPITAL DATA → OPERATIONAL INTELLIGENCE → PROACTIVE DECISIONS → BETTER HEALTHCARE OPERATIONS